MATH 152 **QUIZ 7** NAME _____

Mrs. Bonny Tighe 25 points

11.1 – 11.3 TA _____ Wed 10/26/2022

YOU MAY NOT USE ANY COMPARISON TESTS ON THIS QUIZ

1. a. State the Geometric Series Test

b. State the test for Divergence

c. State the Integral Test

d. State the p-series test

2. Find the limits of the sequences and state if the sequence converges or diverges. Is the sequence bounded? If so, what are the bounds. Determine whether the sequence is increasing, decreasing or not monotonic

a) $a_n = \frac{3^n}{n^3}$

b) Find a_n first

$\left\{\frac{1}{2}, \frac{4}{6}, \frac{7}{10}, \frac{10}{14}, \ldots\right\}$

3. Determine whether the series is convergent or divergent and if it is convergent **find the sum**.

a) $\sum_{k=1}^{\infty} \frac{1}{k^2 + 4k + 3}$

b) $\sum_{k=1}^{\infty} \frac{e^{k+2}}{3^{k+1}}$

64 Express $0.\overline{65}$ as a ratio of integers using infinite series.

5. Use the **Integral test** to determine if the series is divergent or convergent.

$$\sum_{n=1}^{\infty} \frac{e^{1/n}}{n^2}$$

6. a) Estimate the error in using the partial sum S_{20} of the series $\sum_{k=1}^{\infty} \frac{1}{k^4}$

b) What partial sum would be necessary for the above series to have 10^{-4} accuracy?

7. Fill in the blank: a) If $\sum_{k=1}^{\infty} a_k \ converges, \ then \ \lim_{k \to \infty} a_k =$ _____

b) What is the symbol for the 30th partial sum of a series?_____

c) If $\sum_{k=1}^{\infty} a_k \ diverges \ and \ \sum_{k=1}^{\infty} b_k \ converges \ then \ \sum_{k=1}^{\infty} (ca_k + db_k)$ _____

Circle True of False

TRUE FALSE d) If $\lim_{k \to \infty} a_k = 0 \ then \ \sum_{k=1}^{\infty} a_k$ converges.

TRUE FALSE e) The Integral Test can be used to test $\sum_{k=2}^{\infty} \frac{k^2 - k^4}{k^6 - 1}$ for convergence

MATH 152 **QUIZ 7A** NAME _____

Mrs. Bonny Tighe 25 points

11.1 – 11.3 TA _____ Wed 10/26/2022

YOU MAY NOT USE ANY COMPARISON TESTS ON THIS QUIZ

1. a. State the Geometric Series Test

b. State the test for Divergence

c. State the Integral Test

d. State the p-series test

2. Find the limits of the sequences and state if the sequence converges or diverges. Is the sequence bounded? If so, what are the bounds. Determine whether the sequence is increasing, decreasing or not monotonic

a) $a_n = \frac{n^3}{3^n}$

b) Find a_n first $\left\{\frac{1}{2}, \frac{3}{5}, \frac{5}{8}, \frac{7}{11}, \ldots\right\}$

3. Determine whether the series is convergent or divergent and if it is convergent **find the sum**.

a) $\sum_{k=1}^{\infty} \frac{1}{k^2 + 3k + 2}$

b) $\sum_{k=1}^{\infty} \frac{e^{k+1}}{3^{k+2}}$

64 Express $0.\overline{35}$ as a ratio of integers using infinite series.

5. Use the **Integral test** to determine if the series is divergent or convergent.

$$\sum_{n=1}^{\infty} \frac{e^{1/n}}{n^2}$$

6. a) Estimate the error in using the partial sum S_{20} of the series $\sum_{k=1}^{\infty} \frac{1}{k^5}$

b) What partial sum would be necessary for the above series to have 10^{-4} accuracy?

7. Fill in the blank: a) If $\sum_{k=1}^{\infty} a_k \ converges, \ then \ \lim_{k\to\infty} a_k =$ _____

b) What is the symbol for the 30th partial sum of a series?_____

c) If $\sum_{k=1}^{\infty} a_k \ diverges \ and \ \sum_{k=1}^{\infty} b_k \ converges \ then \ \sum_{k=1}^{\infty} (ca_k + db_k)$ _____

Circle True of False

TRUE FALSE d) If $\lim_{k\to\infty} a_k = 0 \ then \ \sum_{k=1}^{\infty} a_k$ converges.

TRUE FALSE e) The Integral Test can be used to test $\sum_{k=2}^{\infty} \frac{k^2 - k^4}{k^6 - 1}$ for convergence